ADAPTIVE SUSTAINABLE TECH-ENABLED ROBOTIC ASSISTANT

PRESENTED BY: Manas Ranjan Das, Chinmayee Moharana, Raj Tripathy, Bhaktishree Vibek

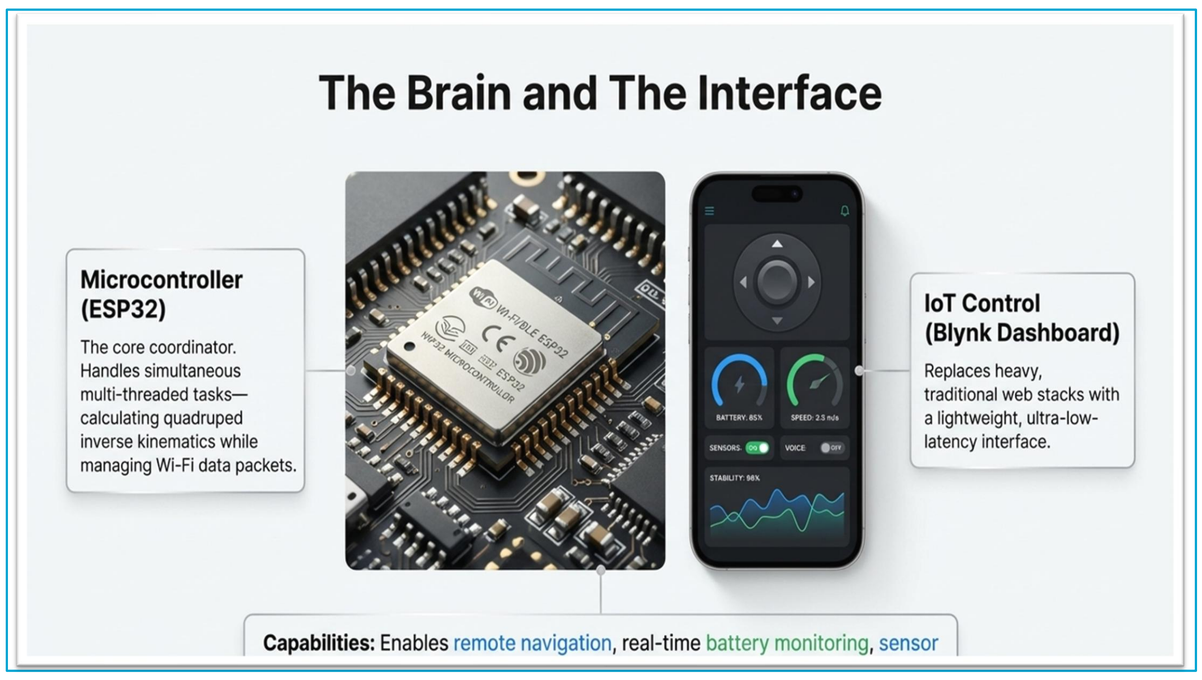
Regd No: 2301206189, 2301206174, 2301206195, 2301206171

Guided by ***Er. Jyoti Ranjan Sahoo***

Abstract

Adaptive Sustainable Tech-enabled Robotic Assistant (A.S.T.R.A.) is a compact quadruped robotic platform designed for stable locomotion and real-world deployment beyond controlled laboratory environments. Built around an ESP32 microcontroller, the system integrates servo-driven leg mechanisms for adaptive movement across uneven terrains. The robot is wirelessly controlled using the Blynk IoT platform, enabling intuitive real-time operation and human-machine interaction. Powered by a Li-ion battery with regulated supply, A.S.T.R.A. ensures reliable and efficient performance. The platform supports scalable integration of sensors for monitoring and navigation, making it suitable for applications such as remote surveillance, industrial inspection, and STEM-based research and education.

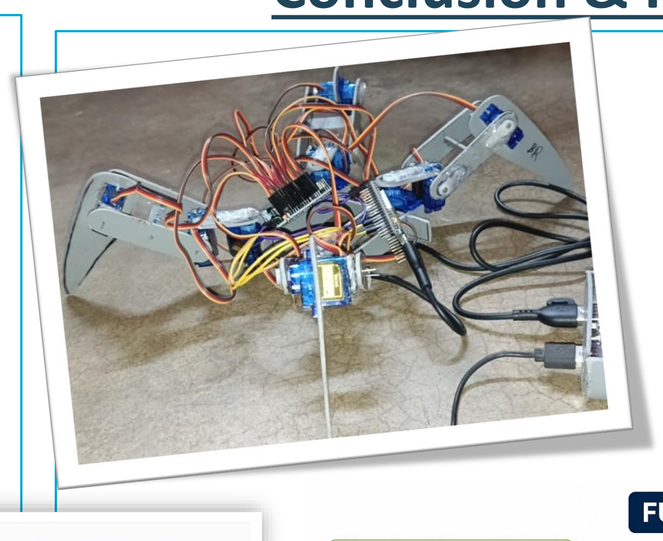
Working principle



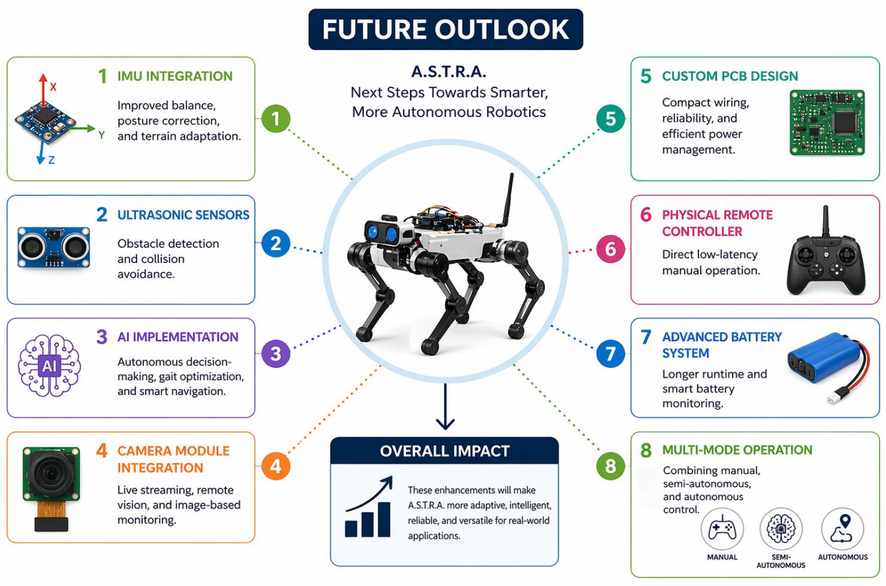
Introduction

Aspect	Details
Problem	Wheeled robots struggle on uneven terrain and depend on centralized charging, limiting real-world deployment.
Gap	Low-cost systems lack intelligent control, real-time monitoring, and adaptability; existing solutions are often complex and expensive.
Need	Scalable, IoT-enabled robotic platforms with intuitive remote control for dynamic environments.
Solution	**A.S.T.R.A.** – a compact quadruped robot designed for stable locomotion across rough terrains.
Core Technology	ESP32-based control, servo-driven legs, and Blynk-based wireless interface for real-time operation.
Outcome	Cost-effective, flexible system suitable for surveillance, inspection, and STEM applications.

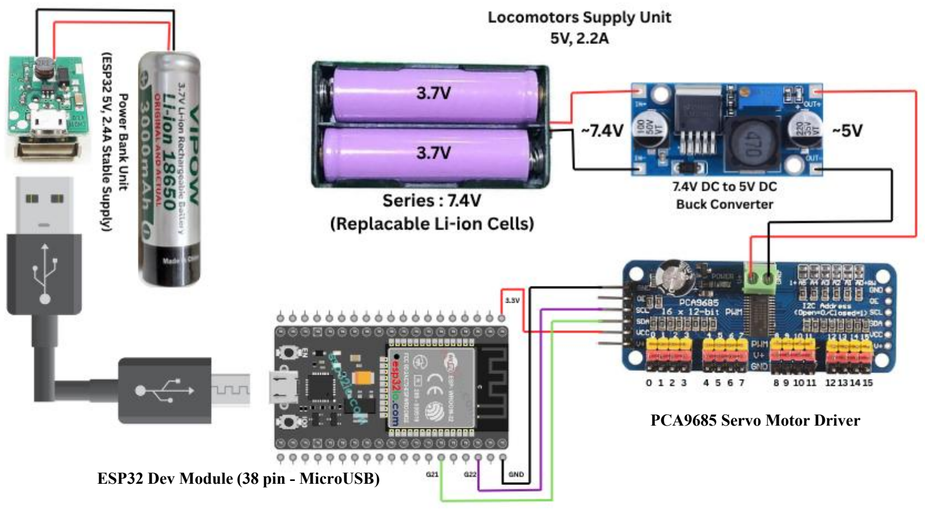
Conclusion & Future scope



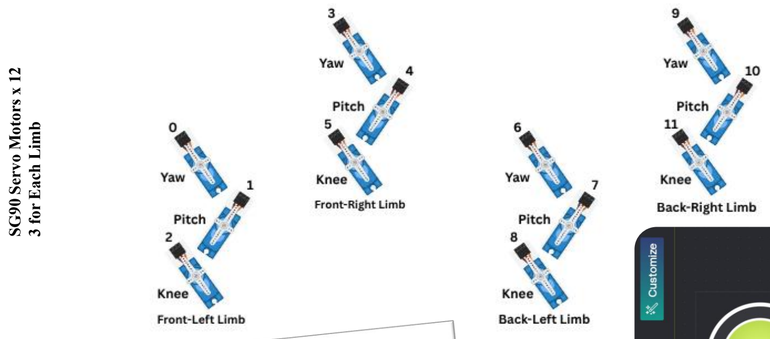
A.S.T.R.A. successfully demonstrates a compact quadruped robotic platform capable of stable locomotion and wireless real-time control. By integrating ESP32 processing, servo-based movement, and a Blynk IoT interface, the system offers a low-cost, scalable, and practical solution for applications beyond controlled laboratory environments such as surveillance, inspection, and STEM learning.

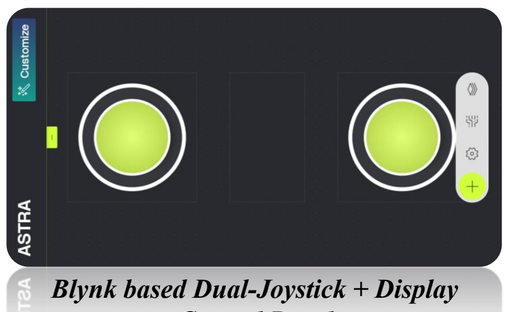


System design

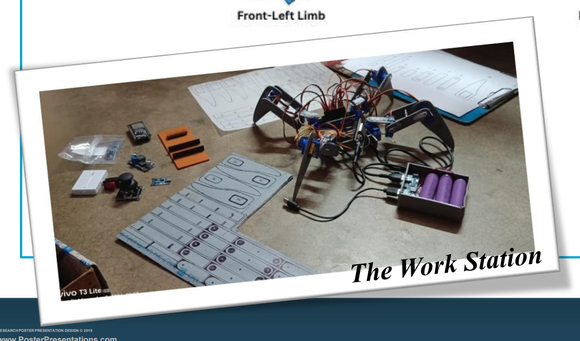


ARM



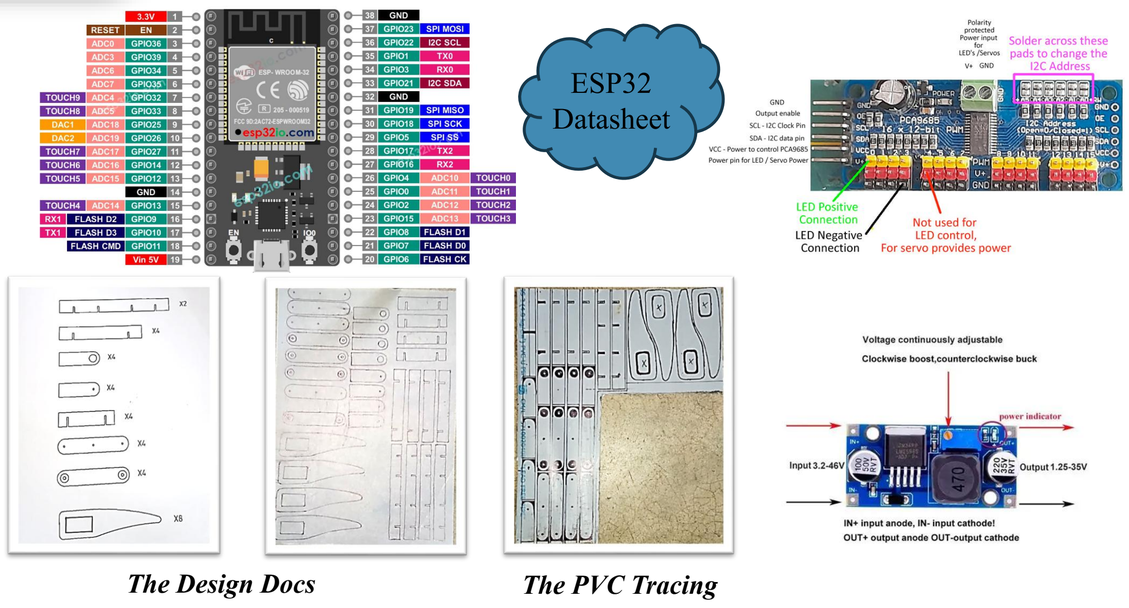


Blynk based Dual-Joystick + Display Control Panel



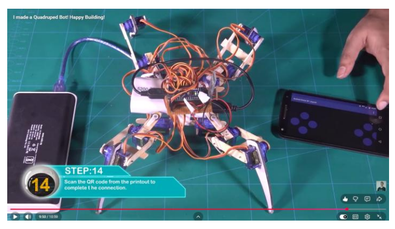
The Work Station

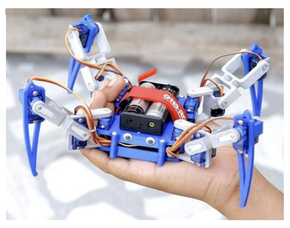
References



The Design Docs

The PVC Tracing





Existing Works were Arduino & Bluetooth based…..

We use ESP32 WIFI + Blynk Cloud

Mechanical Design & Assembly References

1. **SPARKLERS** : We Are The Makers. "Building a Bluetooth Controlled Spider Robot using Arduino."
2. **Telugu Experiments with Lokesh**. "How I Built a 3D Printed Spider Robot | Walking Spider Robot Made from Scratch."
3. **Maker 101**. "How to make Quadruped Spider Robot (using Arduino and 3D parts)."
4. **Skynet Robotics.** "3D printed spider robot using Arduino."
5. **Indian LifeHacker**. "How to make a WALKING SPIDER ROBOT at home | 3D printed crawling robot."
6. **The Wrench**. "How to Make a Spider Robot."
7. **Andreas Lorenz**. "Quadruped Robot."

Theoretical References

1. **Rectenna Circuit:** An Overview of RF Energy Harvesting – Working and Applications
2. **Blynk Dash Boarding:** Introduction | Blynk Documentation
3. **MERN - WebConnect:** MERN Stack – GeeksforGeeks
4. **Quadruped Guide:** Quadruped Robot: A Complete Guide to Quadruped Robots - Humanoid Robotics Technology
5. **4-Legged Quadruped:** (PDF) Quadruped robot - Four Legged Robot
6. **Development of Quadruped Walking Robots:** Development of quadruped walking robots: A review - ScienceDirect

GitHub Repository:
https://github.com/AlienMinus/ASTRA

RESEARCH POSTER PRESENTATION DESIGN © 2015
www.PosterPresentations.com